2024 IEEE 27th International Conference on
Intelligent Transportation Systems (ITSC)
September 24- 27, 2024. Edmonton, Canada

Balancing Individual and Collective Interests: Integrating Transformer-Based PPO and Social Intelligence for Autonomous Vehicles at Unsignalized Intersections

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***Abstract*—Autonomous vehicles (AVs) promise a transformative shift in transportation paradigms, yet their integration into mixed-traffic environments poses complex challenges. In this paper, we propose a novel framework that combines Proximal Policy Optimization (PPO) with the Self-attention mechanism from transformers and Social Value Orientation (SVO) to address the delicate balance between individual and collective interests at unsignalized intersections. By integrating SVO, a psychological concept characterizing individuals' preferences for cooperation and competition, into PPO with Self-attention mechanism, our framework enables AVs to make ethically informed decisions that consider both individual and collective welfare. Transformers, known for their ability to capture long-range dependencies and model complex relationships within data, enhance the reinforcement learning capabilities by providing more effective and context-aware decision-making. We present a detailed formulation of the PPO-Transformer-SVO framework, discussing how SVO influences AV behavior and cooperation strategies at intersections. Through simulation experiments, we demonstrate the effectiveness of our approach in optimizing traffic flow, minimizing collisions, and ensuring collective and individual benefits among road users. Our results highlight the potential of integrating human-inspired social preferences and advanced reinforcement learning techniques into AV decision-making algorithms to enhance the safety, efficiency, and ethicality of future transportation systems.**

***Index Terms*—Autonomous driving, deep reinforcement learning, cooperative driving, social value orientation, transformer**

I. INTRODUCTION

Autonomous vehicles (AVs) have the potential to revolutionize transportation by offering increased safety, efficiency, and convenience [1]. However, ensuring secure and efficient navigation at intersections characterized by heavy traffic and frequent vehicle interactions remains a difficult task for autonomous driving [2], especially when many human-driven vehicles (HVs) display unpredictable actions. Incorrect anticipation of how other vehicles will behave can significantly impact the decision-making process of the autonomous vehicle, even putting its safety at risk. This gets further complex at unsignalized intersections, where the autonomous vehicle has to deal with cars coming from different directions simultaneously. As more vehicles join the mix and their actions affect each other, it becomes more challenging to predict what action they will take next [3].

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To ensure autonomous vehicles can be safely used on a large scale, we need to tackle various technical, social, and ethical issues and incorporate them into the vehicles' decision-making process [4]. As these vehicles become more integrated into society, it is crucial to consider how they will navigate through complex environments, such as unsignalized intersections, in a way that balances individual and collective interests. This is where the integration of DRL and SVO becomes helpful. SVO, is a well-known concept in social psychology that assesses how much emphasis an individual places on their own well-being in comparison to others, which translates into altruistic, prosocial, or egoistic preferences [5].

Reinforcement learning (RL) is a promising direction to handle self-driving tasks, such as deep Q-learning, Soft Actor-Critic and PPO. In this method, the autonomous vehicle learns from experience by maximizing a reward function that assesses its performance in different traffic scenarios. While reinforcement learning algorithms can optimize the reward function, they might not ensure safety and ethical behavior as they overlook potential risks and consequences [6]. Integrating sophisticated decision-making algorithms with human-centered risk assessment and social-behavioral cues is crucial for attaining this objective. For instance, in Ref. [7], a multi-agent reinforcement learning framework (MARL) was proposed in a highway merging scenario, which study models AV maneuver planning as a partially observable stochastic game to derive optimal policies for socially desirable outcomes. The proposed altruistic AVs exhibited improved traffic flow and safety by forming alliances and influencing HV behavior. Furthermore, Ref. [8] demonstrated how introducing SVO into the DRL Reward Function design influences the ego-vehicle strategies, achieving behaviors that range from egoistic to pro-social without affecting pedestrian safety. Similarly, Taghavifar *et al.* presented a hybrid human-centric reinforcement learning framework which is proposed to model the interaction dynamics between an ego car and other road users, specifically bicycles, in a traffic scenario by integrating the SARSA (State-Action-Reward-State-Action) algorithm to learn the optimal policy for the ego vehicle while also incorporating risk cost to estimate the probability of collisions [9]. The study by Tram *et al.* [10], proposed a hierarchical decision-making algorithm for autonomous vehicles at intersections, combining a reinforcement learning decision module with a model predictive control (MPC)

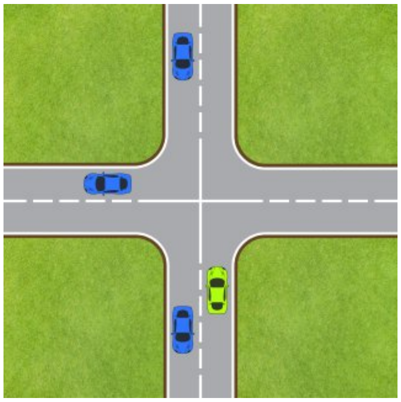


Fig. 1: Illustration of the task scenario.

planner. This approach was limited to scenarios where vehicles travel straight. However, incorporating scenarios where vehicles turn left or right would significantly complicate the problem.

To our knowledge, there is currently a lack of research on the application of SVO at unsignalized intersections, especially when the autonomous vehicle intends to make a left turn in the presence of human-driven vehicles. In our paper, we integrate the SVO framework into transformer-based PPO to shape the reward function and prioritize the comfort and safety of human-driven vehicles and the autonomous vehicle's goals. This approach can promote ethical and responsible decision-making in autonomous driving. The primary contributions of this paper are as follows:

- We develop a modified reward function architecture that incorporates SVO, shaping the decision-making strategies of the ego vehicle. This adjustment allows for a spectrum of behaviors, from egoistic to pro-social, which we demonstrate leads to enhanced traffic flow and increased safety for all road users.
- A Transformer-Based PPO (TPPO) framework is proposed. This novel approach leverages the power of transformers to enhance the decision-making process in complex traffic scenarios, such as unsignalized intersections, by effectively capturing long-range dependencies and interactions between vehicles.
- By utilizing the *Highway-env* simulation package [11], which is built based on the OpenAI Gym Environment [12], we perform simulations at an unsignalized intersection, aiming to identify the optimal SVO setting that promotes altruism, where no vehicle is prioritized over another.

II. PROBLEM FORMULATION

This section starts by presenting the task scenarios addressed in this work. Subsequently, the structure of the learning environment is explained. We are interested in exploring how an AV can be trained to turn left at an unsignalized intersection in the presence of HVs while considering how the ego vehicle can optimize for a social utility that includes the interests of nearby vehicles.

A. Problem Statement

In this scenario, an autonomous vehicle aims to turn left at an unsignalized four-way intersection to reach its destination (see Figure 1). The ego car starts from the lower zone of the intersection, while several HVs start at random speeds from the upper part and approach the intersection, turning right or proceeding straight. The behavior of these HVs is governed by the Intelligent Driver Model (IDM) [13]. For the purposes of this study, we assume that the ego vehicle has access to the position and velocity information of HVs. However, the driving intentions of these vehicles remain unknown to the ego vehicle [3]. The goal is to generate a sequence of actions that allows the ego vehicle to reach the target point without collisions with other HVs, while considering their interests. To reach an efficient and safe traffic flow, we aim to find an optimal SVO value that balances the AV's selflessness and selfishness.

B. Action and Observation Spaces

We use a numeric representation to capture the kinematics of neighboring vehicles in an agent's observation, which is explicitly defined by *Highway-env*. At each time-step t, the agent observes the kinematic features of vehicles at the intersection. Consequently, the state matrix can be defined as follows:

$$s = (s_i)_{i \in [0,N]} \tag{1}$$

where the joint state of road traffic with one ego-vehicle denoted s_0 and N other vehicles can be described by a list of individual vehicle states:

$$s_i = \begin{bmatrix} x_i & y_i & v_i^x & v_i^y & \cos\psi_i & \sin\psi_i \end{bmatrix}^\top \tag{2}$$

where x_i and y_i are the vehicle's current position in the world coordinate system, respectively; v_i^x and v_i^y are the speed of the vehicle among the X-axis and Y-axis, respectively; ψ_i is the heading angle of the vehicle at time-step t. The agent must drive a vehicle by controlling its acceleration chosen from the set of abstract meta-actions in a predefined path from the starting point to the target point:

$$\mathcal{A}_i = [\texttt{Decelerate, Idle, Accelerate}] \tag{3}$$

The discrete meta-action type introduces a layer of speed controllers above the continuous low-level control, allowing the ego-vehicle to follow the road at a specified velocity automatically. The meta-actions available include changing the speed, which acts as set points for the low-level controllers [11]. To provide a concise representation of the proposed framework, Figure 2 is presented.

C. Social Reward Function

The designed reward function is composed of two terms:

$$R = r_{ego} \cos\varphi + r_{hv} \sin\varphi \tag{4}$$

The r_{ego} term accounts for the vehicle's own performance parameters, while r_{hv} captures the intentions and comfort of surrounding vehicles. Additionally, φ represents the ego vehicle's SVO value, which influences the vehicle's behavior toward more altruistic or egoistic tendencies. Notably, r_{ego} itself is an aggregate of multiple terms:

$$r_{ego} = r_c + r_v + r_g \tag{5}$$

The first term penalizes collisions between the ego vehicle and HVs and it is defined as:

$$r_c = \begin{cases} -5 & \text{if collision} \\ 0 & \text{else} \end{cases} \tag{6}$$

r_v is a speed reward aimed at encouraging the AV to complete the task as quickly as possible, and r_g is the reward granted for reaching the designated goal and they are set to 1 and 1 respectively. The second term of Eq.4 is used to capture the intentions and comfort of HVs, and it is defined as follows:

$$r_{hv} = r_j^{\mathrm{HV}} = \sum_j^N (r_s)_j \tag{7}$$

where N is the number of human-driven vehicles, and $(r_s)_j$ denotes the positive reward given to the AV, if the AV keeps a safe distance between itself and the front jth HV. The following is the definition of this reward:

$$r_{hv} = r_s = 0.5 \text{ if } distance > 25 \tag{8}$$

It should be noted that these reward function values were obtained through a trial-and-error process.

III. Deep Reinforcement Learning Methods

A. Proximal Policy Optimization

Proximal Policy Optimization (PPO) is a popular algorithm in the field of reinforcement learning due to its effectiveness and simplicity [14]. PPO is an on-policy algorithm that robustly optimizes a stochastic policy. PPO revolves around two primary concepts: the policy gradient and the clipping mechanism to maintain a stable update. The objective function in the PPO algorithm is shown as follows:

$$L_{clip}(\theta) = \hat{\mathbb{E}}_t \left[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t) \right] \tag{9}$$

where $r_t(\theta)$ represents the probability ratio $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)}$, π_θ is the policy parameterized by θ, a_t is the action taken, and s_t is the state at time t. $\hat{A}_t$ is an estimator of the advantage function at time t, and ϵ is a small constant. The advantage function is defined as follows:

$$\hat{A}_t = \delta_t + (\gamma\lambda)\delta_{t+1} + ... + (\gamma\lambda)^{T-t+1}\delta_{T-1} \tag{10}$$

where $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), r_t$ is the reward at time $t, V(s)$ is the value function at state s, γ is a discount factor and λ helps to balance the bias-variance trade-off in advantage estimation. Furthermore, the clipping function ensures that the ratio $r_t(\theta)$ does not move too far from 1. This helps limit the update steps and makes training more stable. Additionally, Algorithm 1 provides more detailed steps via pseudo-code.

Algorithm 1 Pseudo-code of the proposed PPO-based decision-making for AV with SVO

```
Initialize policy parameters θ
Initialize value function parameters φ
for Episode=1 to N do
    Reset the environment
    Initialize episode buffer
    while not terminated do
        Execute action using the policy: a_t ← π_θ(s_t)
        Observe reward r_t and next state s_{t+1}
        Store transition (s_t, a_t, r_t, s_{t+1}) in the episode buffer
        s_t ← s_{t+1}
    end while
    Compute returns and advantage estimates Â_t from the episode buffer
    Optimize L(θ, θ_old) with respect to θ, using SGD with clipping: ε = 0.2
    Fit value function by regression on mean-squared error: φ ← arg min_φ Σ_t (V_φ(s_t) − R_t)^2
end for
```

B. Transformer Architecture

Transformers have revolutionized various fields such as natural language processing [15] and computer vision due to their ability to model complex dependencies and relationships within data. The self-attention mechanism is a core component of transformers that allows the model to weigh the importance of different elements in the input sequence dynamically. In our framework, we apply self-attention to the state representation of autonomous vehicles to enhance decision-making at unsignalized intersections. To calculate the self-attention, the input state representation $x \in \mathbb{R}^d$ is linearly mapped to three vectors: a query vector $q = xW_q \in \mathbb{R}^{d_q}$, a key vector $k = xW_k \in \mathbb{R}^{d_k}$, and a value vector $v = xW_v \in \mathbb{R}^{d_v}$ with dimensions $d_q = d_k = d_v$. Then, vectors derived from distinct inputs are aggregated into three matrices: Q, K, and V. In our application, the input state representation x consists of features representing the vehicle's dynamics and position (Equation. 2). These features are linearly transformed into the query, key, and value vectors:

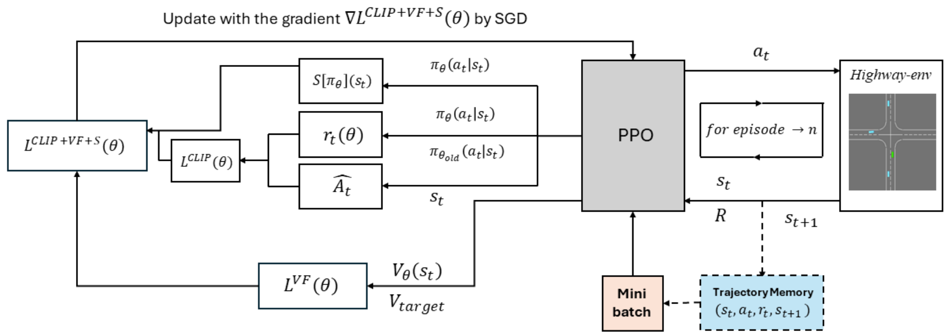


Fig. 2: A general block diagram of the proposed framework based on SVO using PPO.

$$Q = xW_Q \tag{11}$$

$$K = xW_K \tag{12}$$

$$V = xW_V \tag{13}$$

Where W_Q, W_K, and W_V are learned weight matrices. The attention mechanism then calculates the attention scores, normalizes them, and computes the weighted sum of the value vectors:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{14}$$

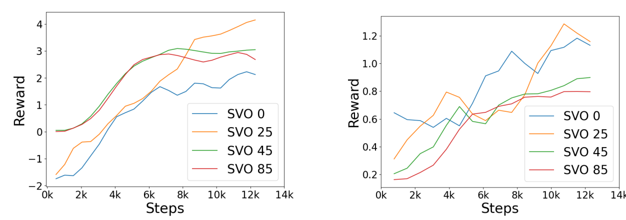
Here, $d_k = 128$, is the dimension of the key vectors, and the softmax function ensures that the attention weights sum to one. This allows the model to focus on relevant parts of the state representation, capturing the complex dependencies between different elements of the state, such as the relationships between the vehicle's position, velocity, and orientation relative to other vehicles.

In our framework, the transformer architecture is integrated with PPO to handle the complex decision-making processes in autonomous driving scenarios. The self-attention mechanism enables the model to effectively process and prioritize information from various traffic participants, leading to more informed and context-aware decisions. By leveraging the strengths of transformers, our approach enhances the ability of autonomous vehicles to navigate unsignalized intersections safely and efficiently, considering both individual and collective interests.

IV. SIMULATION EXPERIMENTS

A. Simulation Setup

In this study, we employed PPO and Transformer-Based PPO (TPPO) as deep reinforcement learning methods while the model uses the AV's SVO to address the delicate balance between individual and collective interests at an unsignalized four-way intersection. We compare these two method's performance with and without SVO and discuss the results regarding the safety and efficiency of the proposed framework. Our experiments utilized the *Highway-env* simulation environment, specifically designed to mimic real-world traffic scenarios. The simulation was conducted with a frequency of 15 Hz to ensure a realistic representation of dynamic traffic patterns. The computational experiments used an NVIDIA GeForce RTX 3050 Ti GPU and an AMD Ryzen 7-5800H CPU @3.20 GHz. The effectiveness of the PPO was influenced by the tuning of hyperparameters, which is demonstrated in Table I.



(a) TPPO (b) PPO

Fig. 3: The change of average reward per episode through the training process for TPPO and PPO policies with different values of SVO.

TABLE I: HYPERPARAMETERS USED IN PPO ALGORITHM.

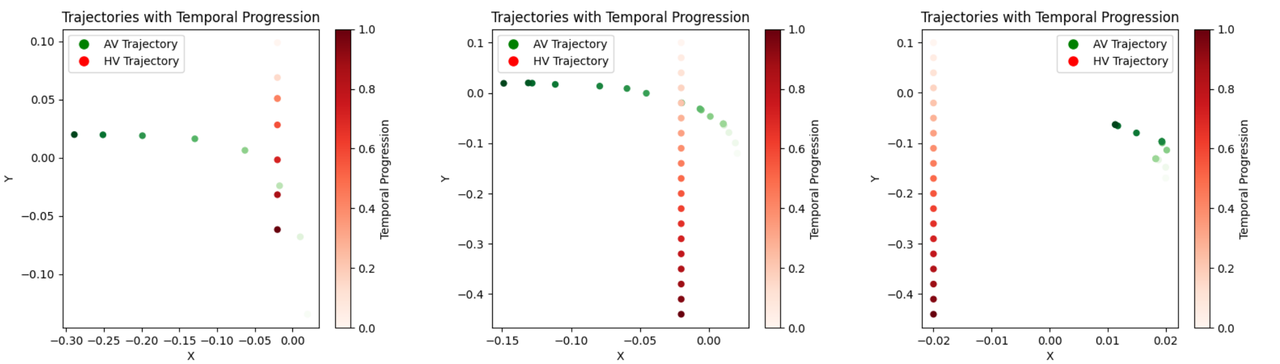
Hyperparameter	**PPO**
N_{steps}	12000
Learning rate (α)	0.005
Discount factor (γ)	0.99
Replay buffer size	-
Batch size	64
Epsilon decay	-
Epochs	10
Clip range	0.2
λ	0.95

B. Training Results

We conducted training with a scenario involving 8 HVs. We ran the simulations for 12,000 steps to train our models using both TPPO and PPO algorithms separately. The goal was to observe and analyze the behavior of autonomous vehicles in an unsignalized four-way intersection with different amounts of SVO. We measured the success rate, collision

TABLE II: COMPARISON OF SUCCESS RATE, COLLISION RATE, AND TIME-OUT RATE AMONG DIFFERENT SVO VALUES.

	$SVO = 0°$			$SVO = 25°$			$SVO = 45°$			$SVO = 85°$		
Methods	succ.(%)	coll.(%)	time-out(%)	succ.(%)	coll.(%)	time-out(%)	succ.(%)	coll.(%)	time-out(%)	succ.(%)	coll.(%)	time-out(%)
TPPO	77	23	0	**98**	**2**	0	40	16	44	1	10	89
PPO	62	28	10	42	12	46	10	4	86	0	4	96



(a) SVO 0 (b) SVO 25 (c) SVO 85

Fig. 4: AV and HV trajectories for three SVO values. Fig. (a)-(c) are generated with TPPO. The temporal progression is illustrated by coloring the AV and HV trajectories from lighter to darker colors. In Fig. (a), AV acts completely selfish, and the HV crosses after the AV has passed, whereas, in Fig. (b) AV yields to the HV, and Fig. (c), AV becomes too cautious and does not complete the turning left task.

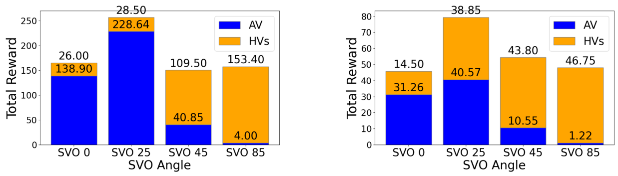
rate, and timeout rate (the instances when an autonomous vehicle fail to reach its destination). These simulations help assess the overall generalization ability of the trained policies with SVO values of 0°, 25°, 45°, and 85° respectively.

Figure 3 shows the change in reward during the training process for both TPPO and PPO. According to these four learning curves for each algorithm, we can find the optimal SVO for the given scenario. For the agent trained with PPO when AV acts with an SVO value of 25°, the agent gets a higher reward than with other values. However, the agent trained with the TPPO algorithm not only demonstrates a strong dependency on SVO angle but also achieves higher overall rewards compared to PPO, particularly with the highest rewards at an SVO of 25°. On the other hand, when SVO is 85°, for both TPPO and PPO, the AV always yields to the HVs and does not reach the target point.

C. Testing Results

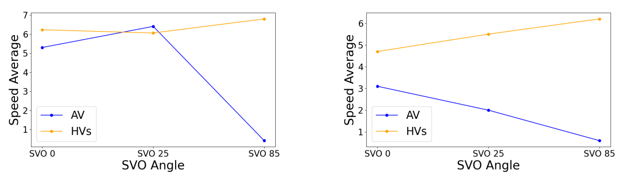
In the testing process, the agent was tested across 100 episodes to assess the robustness of learned behaviors under varying SVO settings. The trajectory of AV and HV, obtained through applying different SVO values, is illustrated in Figure 4, using an agent trained with the TPPO and PPO algorithms. In Figure 4 (a), the AV exhibits egoistic behavior; it accelerates to prevent the HV from crossing, which indicates a low SVO value. Conversely, Figures 4 (b) shows the AV demonstrating altruistic behavior, as it slows down to allow the HV to initiate crossing. Figures 4 (c) presents a scenario where the agent becomes overly cautious at a high SVO value, resulting in the AV failing to complete its task by the time the episode terminates. Table II summarizes the success rate, collision rate, and time-out rate during the testing process with both TPPO and PPO algorithms across a range of Social Value Orientation. The data indicate that as the AV trained with PPO adopts a more egoistic strategy, the success rate improves; however, this also increases the collision rate. Conversely, as the AV adopts more pro-social behavior, the time-out rate increases, mainly because the AV yields to HVs. Remarkably, at the optimal SVO for TPPO, there appears to be a balanced outcome where the success rate does not fall dramatically while simultaneously reducing the collision rate.

The bar graph presented in Figure 5 highlights the total rewards garnered by AV and HVs at varying levels of SVO. Significantly, at an SVO of 25°, the overall reward increases for both policies trained with TPPO and PPO. Moreover, the plot distinctly shows that with TPPO, the overall reward is even higher than with PPO, indicating the superior performance of TPPO in optimizing the rewards for both AV and HVs. Figure 6 displays the relationship between the average speeds of the ego-vehicle and HVs during testing episodes at various SVO settings. For the PPO algorithm, as the SVO value increases, indicating a more altruistic orientation of the AV, the average speed of the HVs consistently rises. This suggests that the AV is yielding more frequently or slowing down to allow HVs to maneuver with less restriction, thus improving the HVs' traffic flow. In other words, the AV's average speed decreases with higher SVO values, which can be attributed to the AV taking a more



(a) TPPO (b) PPO

Fig. 5: HV's reward + AV's reward in the process of 100 testing episodes for TPPO and PPO policies.



(a) TPPO (b) PPO

Fig. 6: The average speed of AV and HVs with different values of SVO.

cautious and considerate approach that prioritizes the flow of HVs over its own speed.

However, for the TPPO algorithm, as the SVO value increases, the average speed of the HVs does not consistently rise. Instead, at an SVO of 25°, the average speeds of both the AV and HVs are close together, suggesting a more balanced interaction between them. This indicates that at an SVO of 25°, both the AV and HVs maintain moderate average speeds, which may suggest an optimal balance for traffic flow efficiency at the intersection where neither is significantly disrupted, aligning with one of the main goals of this study.

V. CONCLUSIONS

In conclusion, this study introduced a novel framework by incorporating SVO, and Transformers into PPO algorithm, to navigate the ethical complexities of autonomous vehicles in an unsignalized four-way intersection. Our research has successfully developed and demonstrated a modified reward function that allows for a continuum of AV behaviors, from egoistic to pro-social, substantially improving traffic flow and safety. Comparative analysis of the TPPO and PPO algorithms, facilitated by the *Highway-env* simulation environment, has shown the benefits of our SVO-adaptive framework. Through testing learned policies, we've established that an optimal SVO value exists where AVs can strike a balance between their own success and the overall welfare of the traffic ecosystem, reducing collision rates without compromising efficiency. Despite the promising outcomes, our study is not without its limitations. The complexity of the scenario is relatively simplified, with HVs entering the unsignalized intersection from a single direction. Future work could enhance realism by introducing HVs from multiple directions, which would more accurately reflect the complexity of real-world traffic scenarios. Another limitation arises from the time-out rates observed when the ego-vehicle adopts an altruistic strategy. There is potential to refine the approach to altruism to strike an even more effective balance that prevents excessive time-outs while maintaining priority for HVs when necessary. Furthermore, we aspire to expand our validation methods in the next phase. It will involve developing a Virtual Reality environment to conduct subjective human factor analysis with real-time human interaction. This human-in-the-loop approach will allow us to gauge human responses to AV behavior more accurately and fine-tune our algorithms accordingly.

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